

Prototype / Demo / Proof of Concept

Proof-of-concept conclusion

Evidence Locker is more than an idea. The repository contains a working full-stack prototype with a React frontend, Express backend, MongoDB models, authentication, complaint workflows, evidence upload and metadata processing, audit and custody support, PDF generation, and an AI chatbot route.

Prototype maturity

Demonstrable product foundation: users can move through the core complaint and evidence lifecycle, while the generated document and chatbot features show how the solution operates end to end.

What this document establishes

- The solution has implemented application surfaces, not only a concept or mockup.
- The main user journey is represented across frontend, backend, storage, and document layers.
- A repeatable demo can show complaint creation, evidence handling, AI guidance, review, and package generation.
- Production deployment still requires hardening and governance, but the technical proof of concept is present.

1. Implemented Prototype Evidence

The current repository demonstrates the following concrete implementation surfaces. Together they establish that Evidence Locker has progressed from an idea to an integrated prototype.

Web application

React/Vite frontend with routing, authentication screens, complaint pages, evidence components, role-aware views, and chatbot UI.

Backend API

Express application with routes for authentication, complaints, evidence, audit, jurisdiction, and chat plus middleware and error handling.

Persistent data model

Mongoose models for users, complaints, evidence, auth sessions, OTP challenges, rate limits, and audit logs.

Evidence pipeline

Multer upload validation, metadata extraction, SHA-256 hashing, GridFS/storage utilities, custody records, and retention support.

Document output

pdf-lib generates structured complaint packages with summaries, evidence references, hashes, custody tables, and a Section 65B draft.

AI assistance

Authenticated Gemini integration supports incident guidance, timeline preparation, evidence-metadata context, and factual report drafting.

Evidence of integration

These are connected capabilities: complaint records link to evidence; evidence links to hashes and custody history; authorized users review status and handoffs; the package generator consumes the structured records; and the chatbot can operate within an authenticated incident context.

2. Demonstrable End-to-End Flow

A practical demonstration can show the following sequence using a test account and synthetic evidence. No real personal or financial data is required for the proof of concept.

- 1 Authenticate**
Register or sign in through the OTP/JWT flow and confirm the user role.
- 2 Create complaint**
Enter victim, location, incident date, fraud category, and category-specific facts.
- 3 Ask the chatbot**
Request help building a timeline or identifying missing documentation; observe bounded guidance.
- 4 Upload evidence**
Upload a synthetic screenshot, document, or transaction record within allowed type and size limits.
- 5 Review metadata**
Observe extracted metadata, manually entered context, file details, and SHA-256 hash.
- 6 Track workflow**
Review status, custody events, audit activity, and role-appropriate actions.
- 7 Generate package**
Create a PDF complaint package containing the structured record and evidence references.

Demo safety rule

Use synthetic complaints and non-sensitive test files during demonstrations. Never expose real OTPs, API keys, private credentials, or unnecessary personal information.

3. Proof-of-Concept Acceptance Checks

The prototype can be assessed through observable behaviors rather than presentation claims. The following checks establish whether the core solution is functioning.

User and access flow	A user can authenticate, access permitted screens, and is prevented from accessing unauthorized complaint records.
Complaint flow	Required fields and fraud-specific fields are validated; complaint status and ownership are persisted.
Evidence flow	Allowed files upload successfully; unsupported types and oversized files are rejected with clear errors.
Metadata and integrity	Metadata is extracted where available, manual context is accepted, and a SHA-256 hash is recorded.
Audit and custody	Evidence and workflow actions create visible audit or custody records for authorized review.
AI chatbot	Authenticated users receive bounded assistance; missing configuration or upstream failure produces an explicit error.
PDF generation	A structured package is generated with complaint details, evidence references, hashes, and custody information.

Passing these checks demonstrates a functioning integrated prototype. It does not, by itself, establish production security, legal acceptance, or institutional readiness.

4. Prototype Scope and Maturity Boundaries

The proof of concept demonstrates the core product direction while making clear what remains before broad production deployment.

Demonstrated now

- Integrated complaint, evidence, metadata, hashing, audit, custody, status, and package-generation workflow.
- Role-aware access for victims, support users, police reviewers, and administrators.
- Authenticated AI chatbot with explicit safety boundaries and evidence-metadata context.
- Repository structure and local development scripts sufficient for continued implementation and testing.

Still required for production

- Automated test coverage, security testing, penetration testing, dependency scanning, and performance testing.
- Production-grade secrets management, monitoring, backups, restore validation, retention enforcement, and incident response.
- Formal privacy, legal, institutional, and jurisdiction reviews before official workflow integration.
- Operational training, support ownership, accessibility validation, and controlled user rollout.

Proof-of-concept judgment

The solution is demonstrably more than a concept: it has implemented user journeys, data models, service routes, evidence processing, document output, and AI assistance. Its maturity is best described as an integrated prototype requiring production hardening, not as a finished public infrastructure service.